MATH3030 Project Plan: Elliptic Curves over Finite Fields and Cryptography

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1 Project

Elliptic Curves over Finite Fields and Cryptography

2 Overview

Elliptic curve cryptography uses the group structure of points on an elliptic curve over a finite field. Since the unit has covered elliptic curves, I'll extend that material to the finite-field setting, explain scalar multiplication, and show how the elliptic curve discrete logarithm problem is used in Diffie-Hellman key exchange.

3 Required reading

The unit has introduced elliptic curves, so I'll revise the group law, point addition and doubling, and the point at infinity. The new material I'll need is finite fields, modular inverses, scalar multiplication, the elliptic curve discrete logarithm problem, and elliptic curve Diffie-Hellman. I'll work through the relevant elliptic curve cryptography sections in Trappe and Washington in week 8 and write up the finite-field definitions at the same time.

4 Sources

Trappe & Washington, Introduction to Cryptography with Coding Theory, sections on public-key cryptography, discrete logarithms, and elliptic curve cryptography. McKean & Moll, Elliptic Curves: Function Theory, Geometry, Arithmetic, chapter 2 for addition on cubics and chapter 7 for arithmetic of elliptic curves. Mao, Modern Cryptography: Theory and Practice, for general public-key cryptography context. Alternative: lecture notes and online sources for cross-checking computations, not for exposition.

5 What I'll do

My report will define elliptic curves in Weierstrass form, explain the non-singularity condition, and derive the point addition and point doubling formulas. Then I'll move to curves over finite fields, compute examples using modular arithmetic, and explain scalar multiplication. I'll state the elliptic curve discrete logarithm problem and apply it to elliptic curve Diffie-Hellman. I'll discuss limitations such as bad subgroup choices and Shor's algorithm, but not try to design a new cryptosystem.

6 Original contribution

Construct a complete toy elliptic curve Diffie-Hellman exchange over a small finite field, probably using a curve such as:

$$y^2 \equiv x^3 + 2x + 2 \pmod{17}.$$

I'll compute the subgroup generated by a chosen base point, choose private keys for Alice and Bob, compute their public keys, verify that both get the same shared secret, and explain why the toy example is insecure. Backup: if the full point set is too long, work only inside the subgroup generated by the base point.

7 Timeline (30 hours total; 5 hrs per week)

W8: Read finite fields and elliptic curve cryptography; choose the toy curve; write the definitions section. Submit this plan.

W9: Revise and derive the group law; write point addition and point doubling section; submit this section for feedback.

W10: Act on W9 feedback; compute the point set or subgroup for the chosen curve.

W11: Write the finite-field, scalar multiplication, and discrete logarithm sections; first full draft for feedback.

W12: Original contribution; complete the toy Diffie-Hellman exchange; revise on feedback.

W13: Polish, check computations, references, submit.

8 Plan revision

I'll revisit at the end of weeks 10 and 12. I may change the curve, finite field, or base point depending on whether the subgroup order is suitable and whether the computations are manageable. I may also shorten the limitations section if it starts distracting from the main focus on elliptic curve Diffie-Hellman.

9 Why this works

- Sources are named with clear roles, including a main cryptography source and a separate mathematical background source.
- Hours are budgeted per week, with an assumed equal distribution.
- The plan commits to a specific cryptographic application: elliptic curve Diffie-Hellman.
- The original contribution has a backup, so the project remains achievable if the point computations become too long.
- The report builds on the unit's elliptic curve content instead of repeating it from scratch.
- Two feedback points are included, with the first one early enough to catch problems in the group law section.
- Plan-revision dates are named, and the parts most likely to change are identified in advance.